

DSA Day 7: Single Linked List - insert/delete/reverse

Book: Data Structures Through C in Depth - Ch3 §3.1.3-3.1.6 · pp 53-62 | Code in Python

Overview

Insertion and deletion are about re-linking pointers in the right order; reversal flips every next pointer.

Key concepts to master

- Insert at begin/end/position
- Delete by value/position
- Reverse by relinking
- Handle the head specially

Python implementation

```
def push_front(head, v):  
    n=Node(v); n.next=head; return n  
def reverse(head):  
    prev=None  
    while head:  
        nxt=head.next; head.next=prev; prev=head; head=nxt  
    return prev
```

Complexity

Insert/delete $O(1)$ at a known node, $O(n)$ to find; reverse $O(n)$.

Common pitfalls

- Wrong order of pointer updates loses the rest of the list
- Not returning the new head

Practice

- Insert at position 3
- Reverse the whole list

